

MILK10k — Exploratory Data Analysis

Skin lesion classification · Individual course project

Student	Stanislaus Lattorff
Deliverable	Session 1 — EDA, project structure, clinical task
Dataset	MILK10k (ISIC Archive) — 10,480 images / 5,240 lesions
Editor / IDE	Visual Studio Code
Environment	Python 3.14 · pandas · NumPy · matplotlib · Pillow

PUBLIC GITHUB REPOSITORY

<https://github.com/stansilaisjfrnep/milk10k-computer-vision>

What is in the repository

Item	Location
EDA notebook (executed, all outputs saved)	notebooks/01_eda_milk10k.ipynb
Reusable analysis package	src/milk10k/
Clinical task explanation	docs/clinical_task.md
Generated numeric summary	reports/eda_summary.md
15 figures + summary tables	reports/figures/, reports/tables/
One-command reproduction	scripts/download_data.sh, scripts/run_eda.py

Headline result of the EDA. MILK10k is **69.4% malignant** — the reverse of the general population — because it only contains lesions a dermatologist already thought were worth biopsying. Every lesion appears **exactly twice** (one dermoscopic, one clinical image), so a random row-level train/test split leaks all 5,240 lesions. These two facts dictate the entire pipeline design.

1 · The clinical task we want to solve

The medical problem

Skin cancer outcomes depend almost entirely on **when** the lesion is found. A melanoma caught while still confined to the epidermis is cured by a simple excision; the same tumour found later has a dramatically worse prognosis. The bottleneck is not treatment — it is **triage**. Far more lesions look worrying than turn out to be cancer, and the specialists who can tell the difference are scarce.

A dermatologist examining a suspicious lesion decides: reassure, monitor, or excise and send to pathology. That decision is made mostly by eye, usually with a **dermatoscope** — a handheld lens pressed against the skin under polarised light, which reveals pigment networks and vessels invisible to the naked eye.

What MILK10k captures

MILK10k is built from exactly that moment. Every one of its **5,240 lesions was photographed twice**, and **95.7% were confirmed by histopathology** — a pathologist examined the excised tissue. The labels are therefore unusually trustworthy.

Modality	What it is	Why it matters
Clinical close-up	Ordinary photograph, visible light	Shows size, borders, surrounding skin
Dermoscopic	Through a dermatoscope, polarised, magnified	Shows sub-surface structure — pigment networks, vessels

The prediction task

Primary target — diagnosis_1, three classes. Input is one or both images of a lesion; output is one label. It is classification: no bounding box, no mask.

Class	Lesions	Share	Clinical meaning
Malignant	3,634	69.4%	Cancer — needs excision
Benign	1,483	28.3%	Harmless — reassure or monitor
Indeterminate	123	2.3%	Pathology itself was inconclusive

Stretch target — 11 diagnosis codes (training_gt.csv): BCC, NV, BKL, SCCKA, MEL, AKIEC, DF, INF, VASC, BEN_OTH, MAL_OTH.

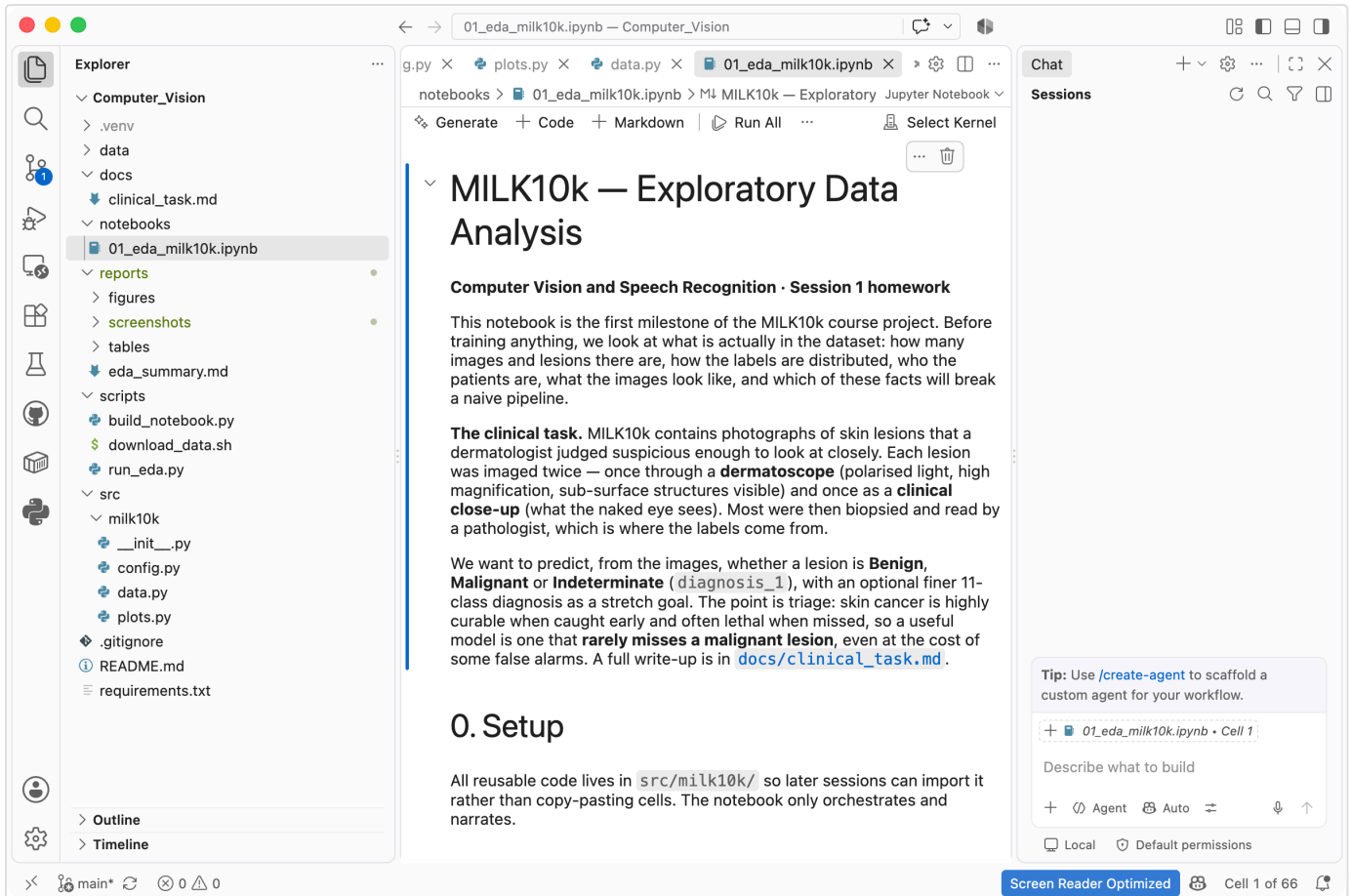
What "good" means here

The two errors are not comparable. A **false negative** sends a patient home with a growing cancer. A **false positive** costs an unnecessary excision — a scar and some anxiety. So the metric that matters is **recall (sensitivity) on malignant lesions**, supported by balanced accuracy, macro-F1, per-class recall and per-skin-tone recall.

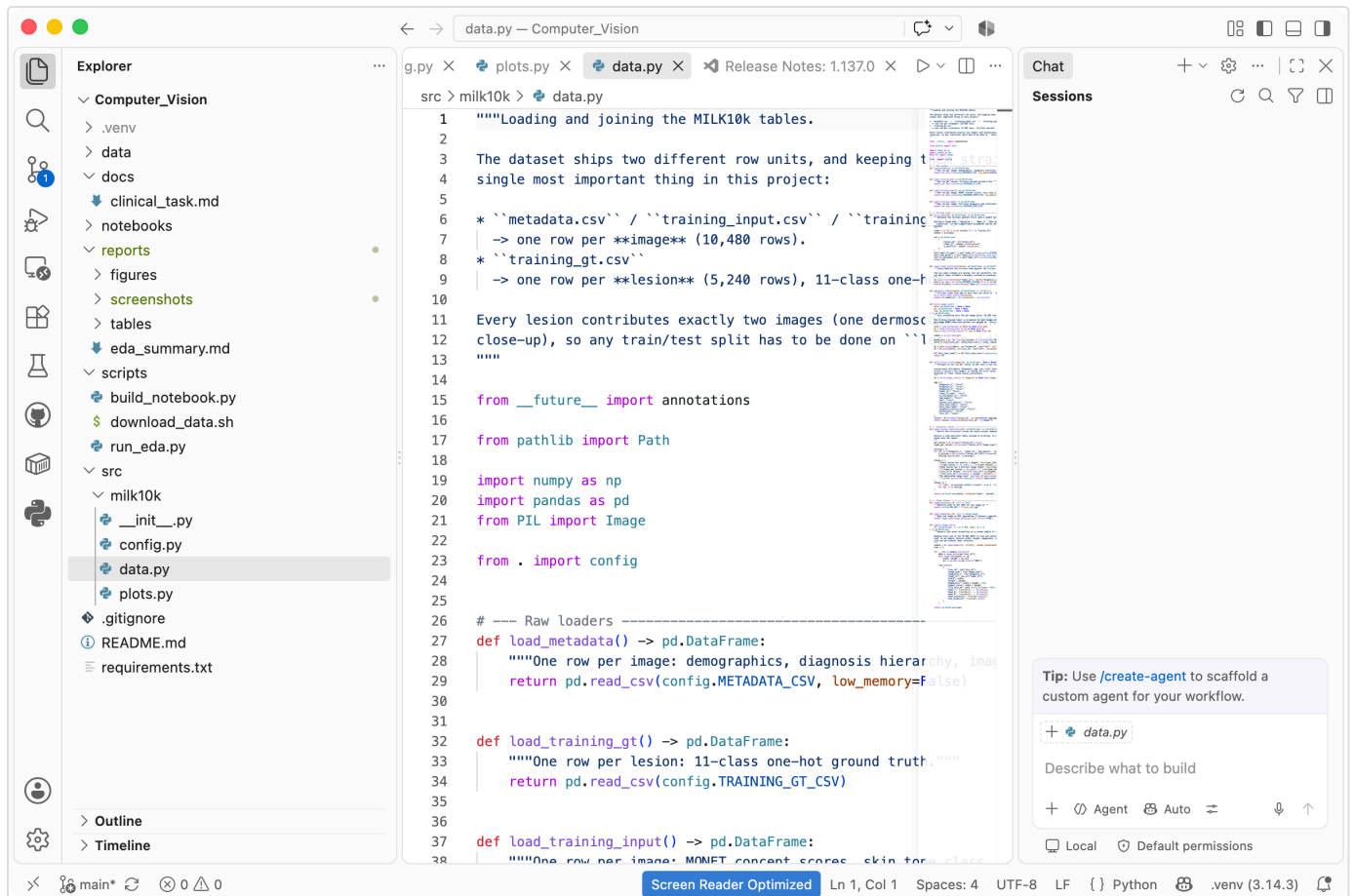
The honest framing. Because the dataset only contains biopsy-referred lesions, a model trained here does not answer "is this skin cancer?" It answers "given that a dermatologist was already worried enough to biopsy this, is it cancer?" Deployed as a consumer app, its positive predictive value would collapse. The realistic role is **decision support** — flagging lesions for a specialist — not diagnosis.

2 · Code editor and project structure

IDE Visual Studio Code, with the project open and the Python environment (.venv) selected. Screenshot of the working repository:



Visual Studio Code — full project tree on the left, the executed EDA notebook open in the editor.



The reusable analysis package `src/milk10k/` — loaders, integrity checks and plotting live here rather than in notebook cells. Status bar: 0 errors, 0 warnings, branch `main`, interpreter `.venv (3.14.3)`.

3 · Repository layout

The design rule: **analysis logic lives in `src/milk10k/`, not in notebook cells**, so later sessions import the same loaders and the same split logic instead of copy-pasting.

```
Computer_Vision/
├── README.md           Project overview + Session 1 findings
├── requirements.txt
├── .gitignore          Excludes data/ (345 MB) and .venv/
├── data/              Dataset – NOT committed (CC BY-NC)
│   ├── milk10k/
│   │   ├── metadata.csv 10,480 rows – one per IMAGE
│   │   ├── images/     10,480 JPEGs
│   │   └── supplements/
│   │       ├── training_gt.csv 5,240 rows – one per LESION (11-class)
│   │       ├── training_input.csv MONET concepts, skin tone, site
│   │       └── training_supp.csv Full-text diagnosis
│   ├── src/milk10k/     Reusable package
│   │   ├── config.py   Paths, label schemes, plot style
│   │   ├── data.py     Loading, joins, integrity checks
│   │   └── plots.py     Every figure in the report
│   ├── notebooks/
│   │   └── 01_eda_milk10k.ipynb ← Session 1 deliverable (executed)
│   ├── scripts/
│   │   ├── download_data.sh Fetch + unzip the dataset
│   │   ├── run_eda.py      Regenerate all figures headlessly
│   │   └── build_notebook.py
│   ├── docs/
│   │   └── clinical_task.md The clinical problem, explained
│   └── reports/
│       ├── eda_summary.md  Generated numeric summary
│       ├── figures/        15 PNGs
│       └── tables/         Summary CSVs
```

Reproducing the analysis

```
git clone https://github.com/stansilaisijfrnep/milk10k-computer-vision
cd milk10k-computer-vision
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
bash scripts/download_data.sh # ~345 MB, from the official ISIC source
python scripts/run_eda.py     # regenerates all 15 figures + tables
```

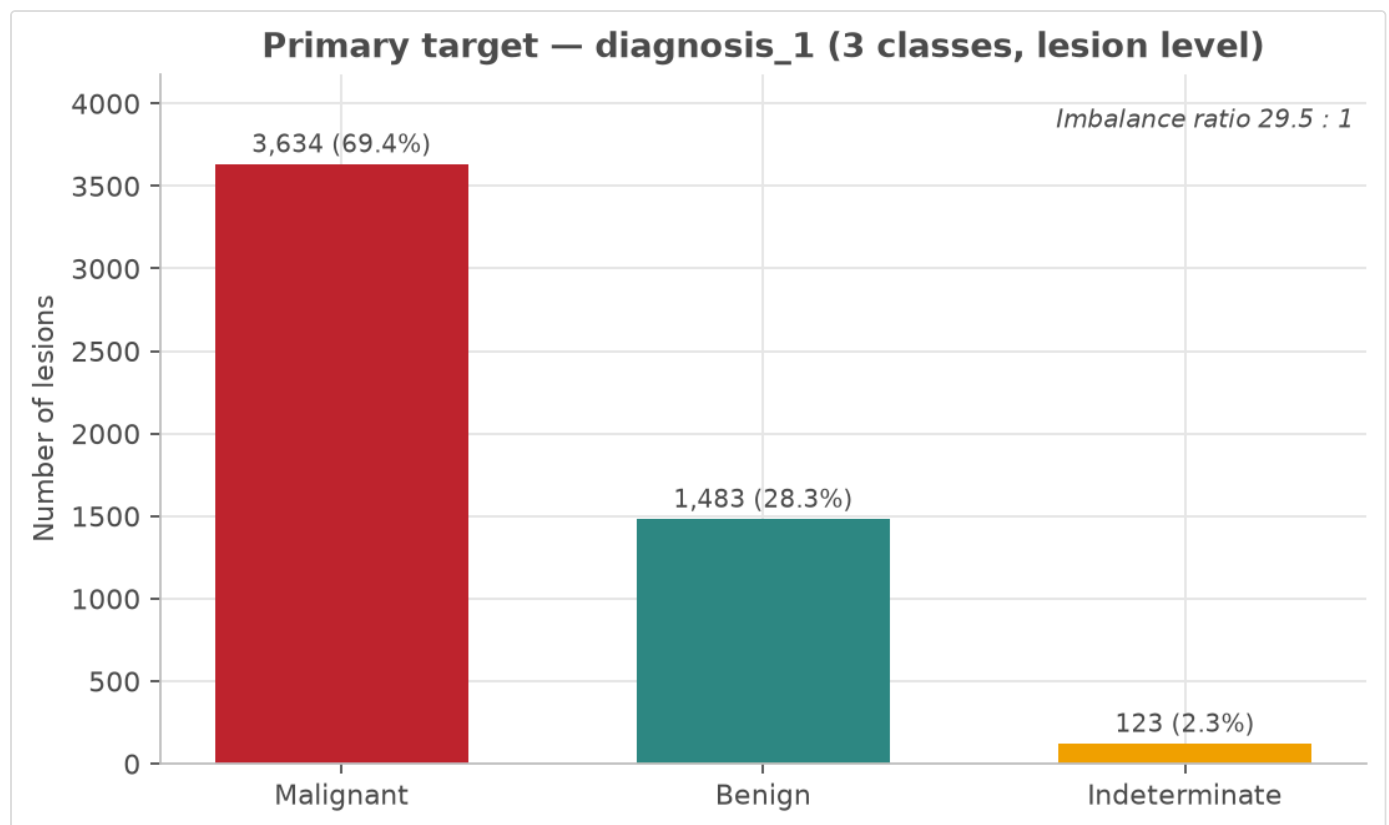
4 · Dataset structure — two different row units

This is the detail that governs everything downstream, and it was verified from the data rather than assumed.

File	Row unit	Rows	Contents
metadata.csv	one image	10,480	Demographics, 4-level diagnosis hierarchy, image type
training_gt.csv	one lesion	5,240	11-class one-hot ground truth
training_input.csv	one image	10,480	7 MONET concept scores, skin tone class, body site
training_supp.csv	one image	10,480	Full-text diagnosis, confirmation type

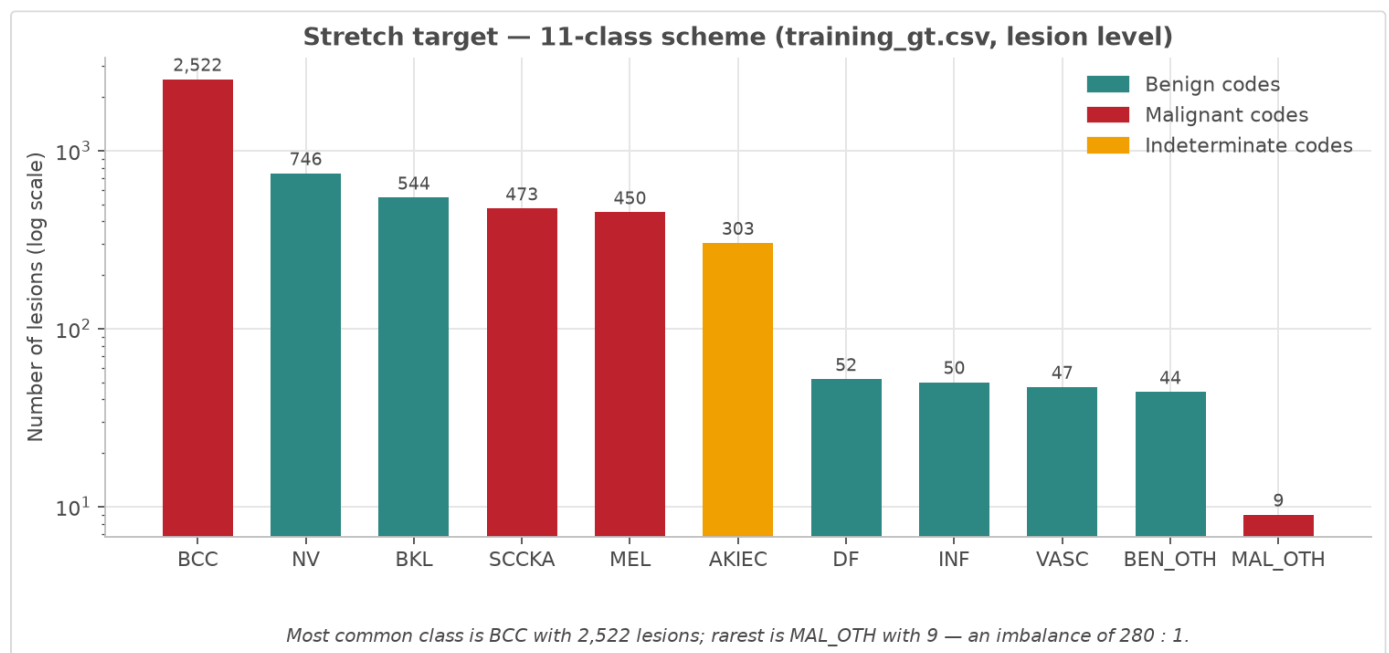
Verified structural checks (all passed): every lesion has exactly 2 images — 5,240 / 5,240; each lesion has exactly one dermoscopic and one clinical image; `isic_id` is unique across all 10,480 rows; diagnosis, age and sex are constant within a lesion; and the 11-class ground truth is genuinely single-label (0 lesions with anything other than exactly one positive class).

5 · EDA — class distribution



Primary target: diagnosis_1 at the lesion level. Malignant dominates — the opposite of the general population.

A model that predicts **Malignant** for every lesion scores **69.4% accuracy** while catching zero benign lesions. This is why accuracy cannot be the headline metric.



Stretch target: the 11-class scheme, log scale. BCC has 2,522 lesions; MAL_OTH has 9.

The 11-class imbalance is **280:1**. Four classes have fewer than 55 lesions, which after a test split leaves only a handful of examples each.

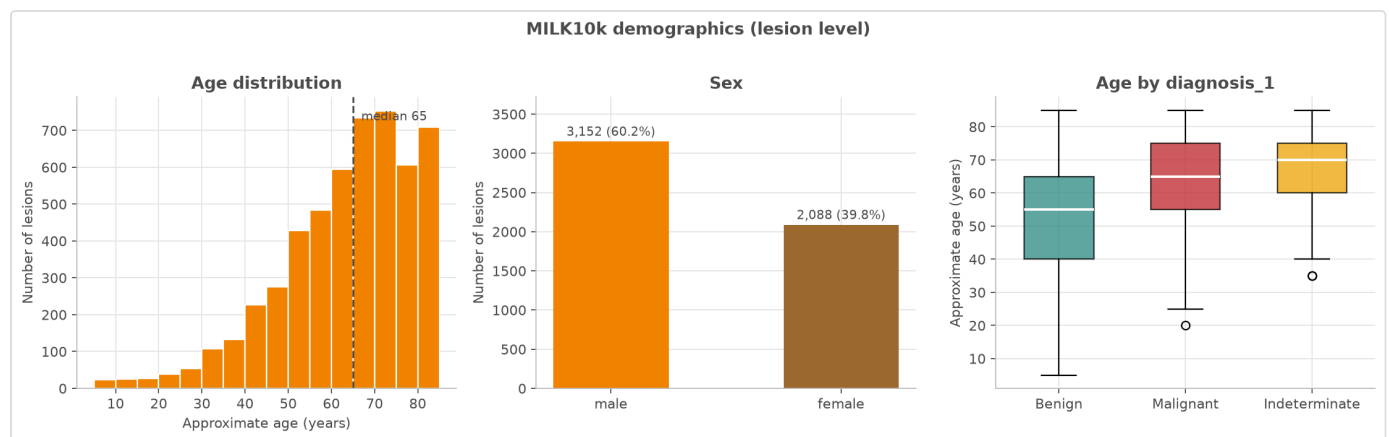
6 · EDA — are the two label schemes consistent?



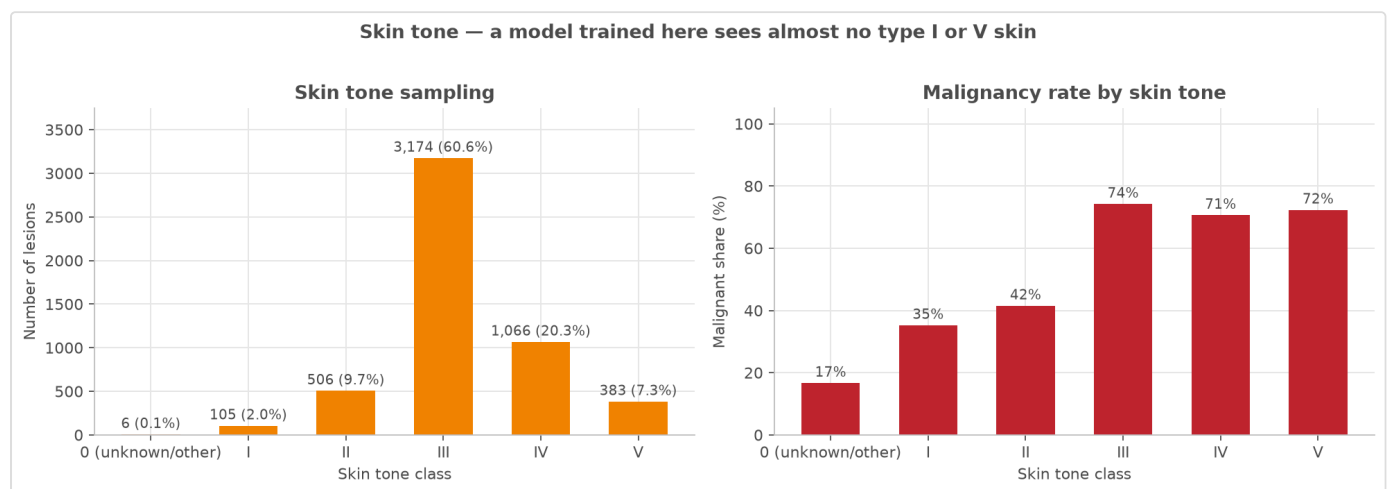
The two label schemes are nested — but not perfectly.

A finding worth flagging. Ten of the eleven codes map cleanly to one value of `diagnosis_1`. **AKIEC does not** — it splits into 123 *Indeterminate* and 180 *Malignant* lesions. Clinically this makes sense (actinic keratosis and intraepithelial carcinoma lie on a continuum), but it means `diagnosis_1` **cannot be derived from the 11-class code by a lookup table**.

7 · EDA — who and what is in the data



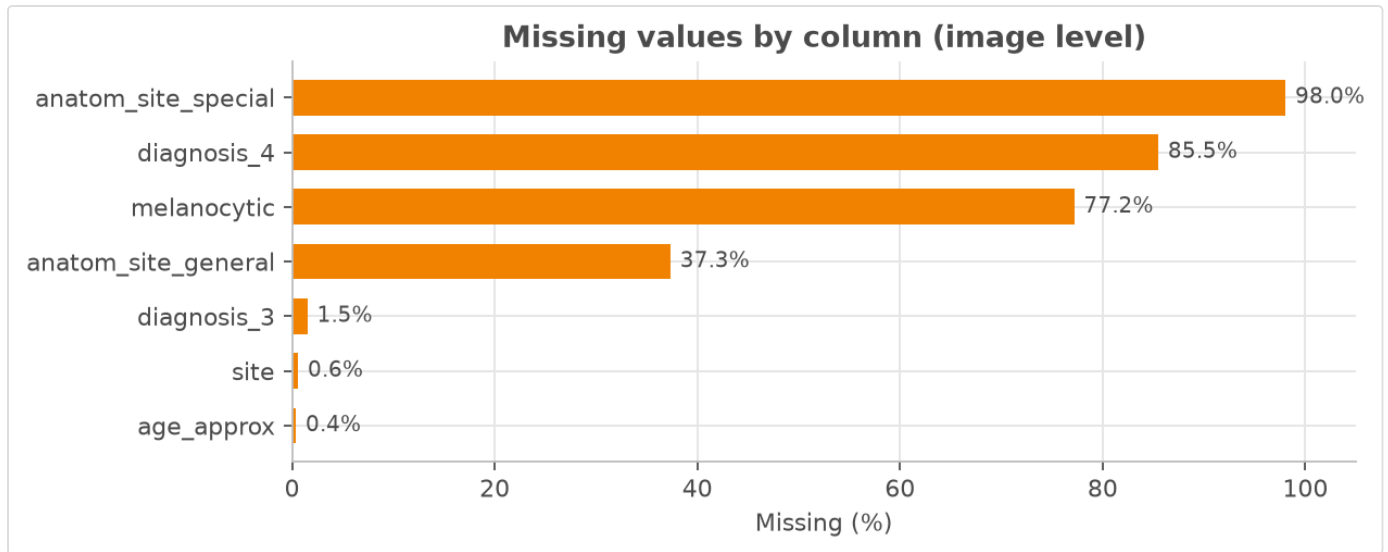
Median age 65, 60% male — a specialist skin-cancer clinic population, not the general public.



Skin tone sampling and malignancy rate. 61% of lesions are a single skin tone class.

Fairness. Skin tone class III alone is **60.6%** of lesions; the lightest (2.0%) and darkest (7.3%) tones are barely present. Aggregate metrics will be dominated by types III–IV, so **per-group recall must be reported separately**. The apparent rise in malignancy with darker tone classes is **confounded** by referral patterns and body site — it is not a causal effect, and with only 105 type-I lesions the leftmost bars are noisy.

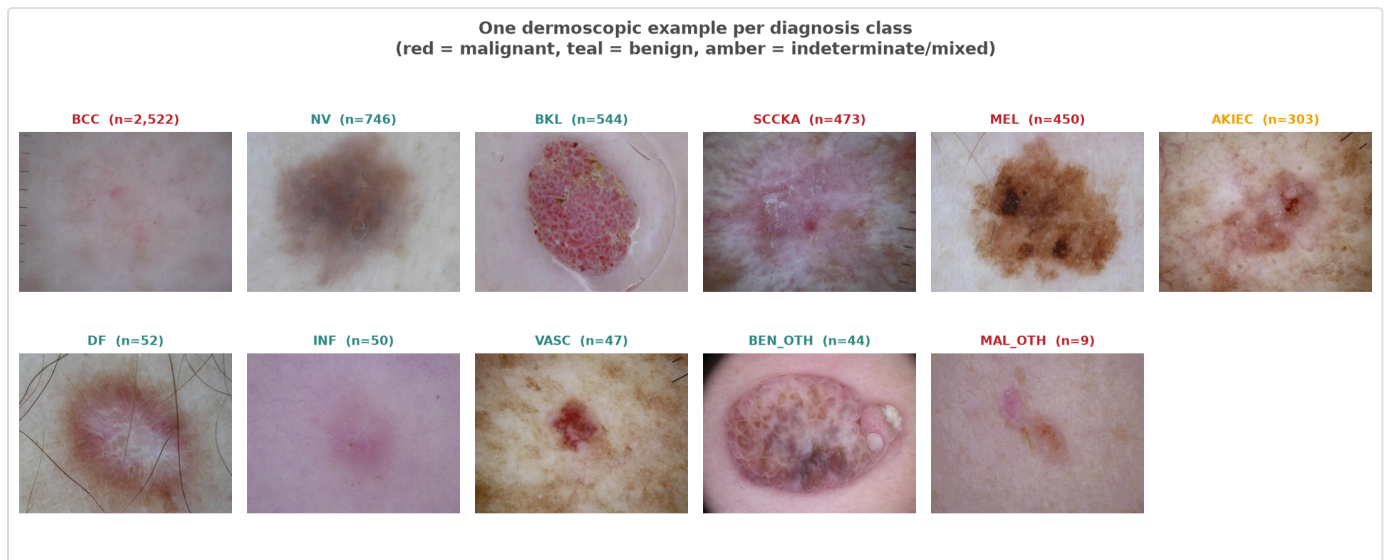
8 · EDA — missing values



Missingness is structural, not random.

melanocytic is blank 77% of the time but is *never* False — it is an annotation, not a boolean, so treating blank as "not melanocytic" would be wrong.

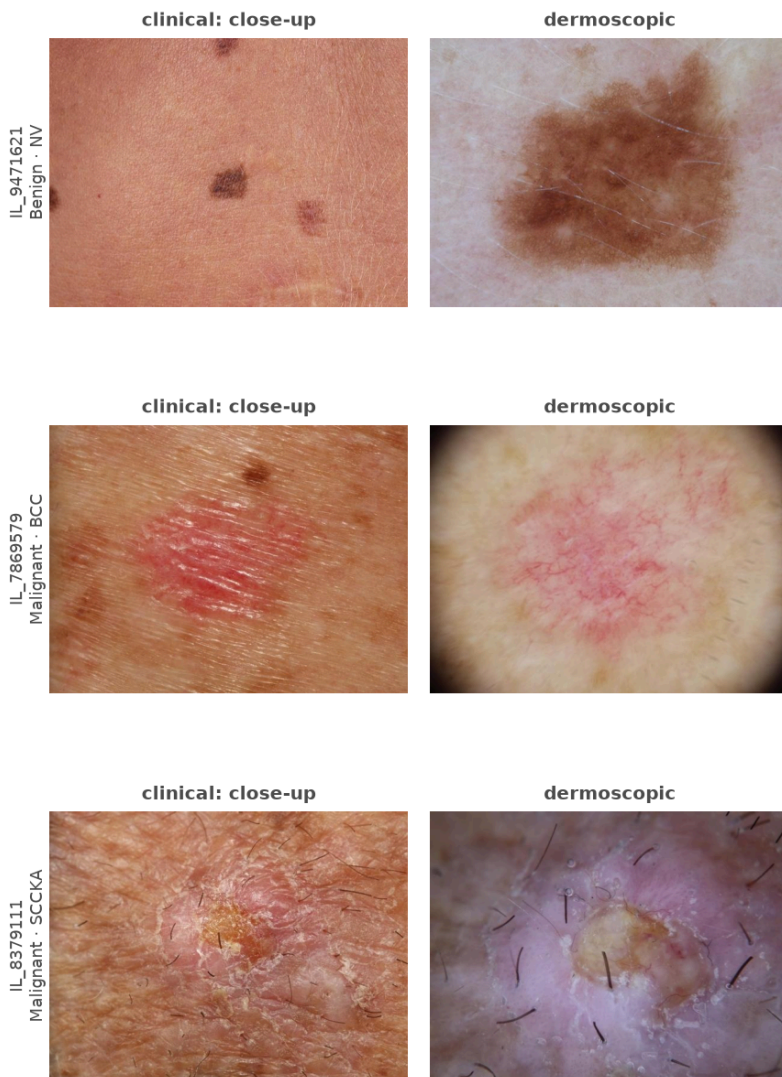
9 · EDA — the images themselves



One dermoscopic example per diagnosis class (red = malignant, teal = benign, amber = indeterminate).

Note how little separates these visually: BCC, the majority class, is often a barely-pigmented pink patch, while a benign nevus is frequently the more dramatic-looking lesion. Colour and size heuristics will not work.

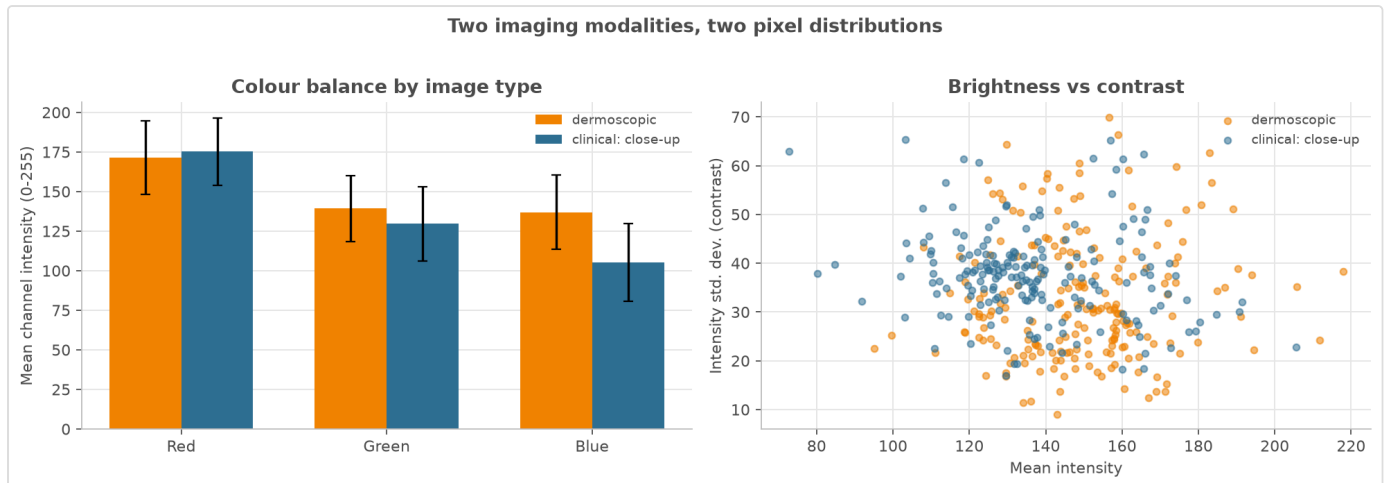
Two images, one lesion — group by lesion_id when splitting



Two images, one lesion — the most consequential structural fact in the dataset.

Why the split must group by lesion_id. If a random row split puts the dermoscopic image in train and the clinical image of the same lesion in test, the model has effectively already seen the test case. Scores rise; real performance does not.

10 · EDA — two imaging modalities



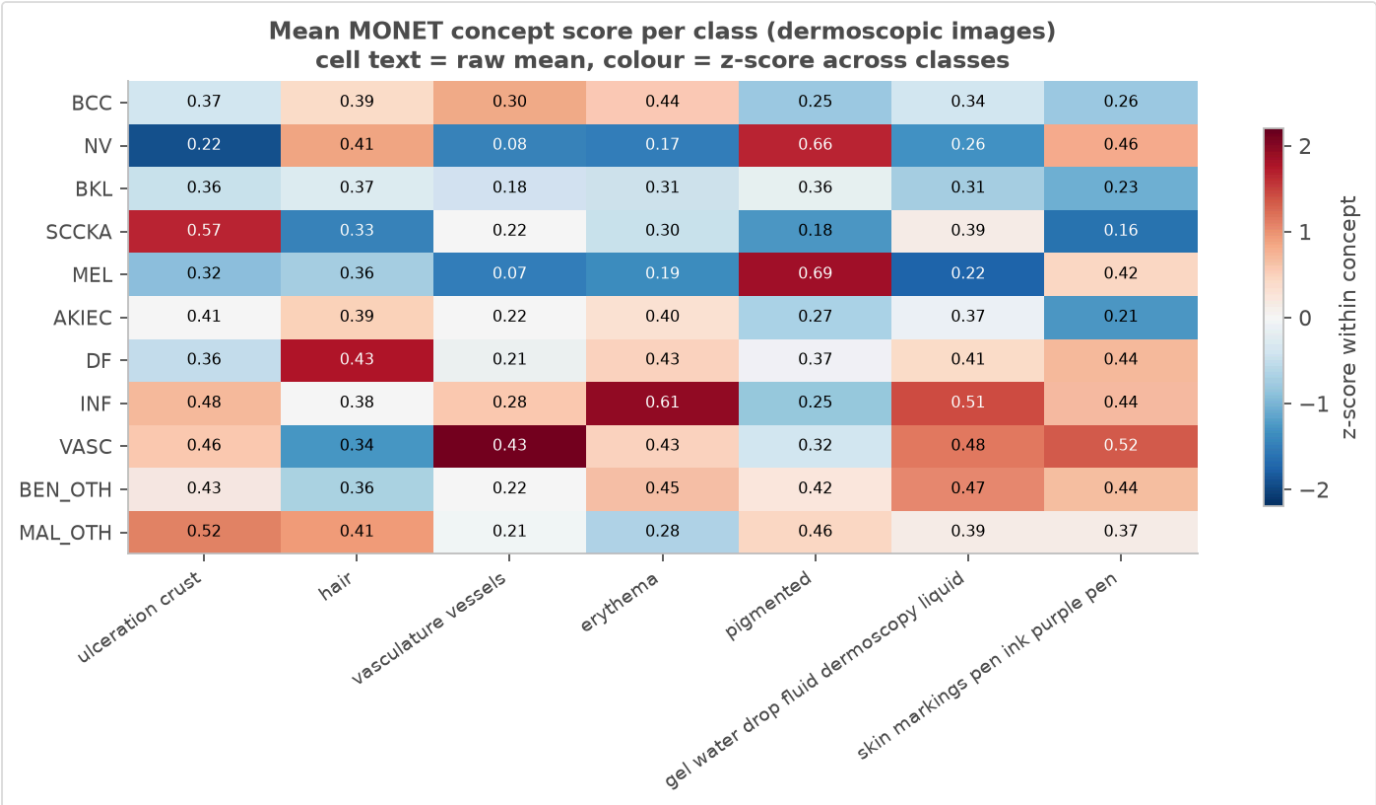
Dermoscopic and clinical images have measurably different pixel statistics.

They are two modalities, not two samples of one. Images also vary in resolution, so a resize/crop step is mandatory before batching.

11 · What the EDA tells us to do next

#	Finding	Consequence for the pipeline
1	Every lesion has exactly 2 images (verified 5,240/5,240)	Use a grouped split on lesion_id . A random row split leaks every lesion.
2	69.4% of lesions are Malignant	Never report bare accuracy. Lead with malignant recall , plus balanced accuracy and macro-F1.
3	11-class imbalance is 280:1	Class weighting, oversampling, or merge the rare tail. Report per-class recall.
4	Two modalities with different pixel statistics	Treat <code>image_type</code> explicitly — stratify, condition, or use two heads.
5	AKIEC straddles Indeterminate and Malignant	Pick one target explicitly; do not derive the 3-class label from the 11-class code.
6	Site missing 37%, melanocytic missing 77%	Metadata needs an explicit "unknown" category, not mode imputation.
7	61% of lesions are one skin tone class	Report per-skin-tone metrics; aggregates hide group failures.
8	Images vary in resolution	Resize/crop is mandatory; document the target resolution.
9	Dataset is biopsy-referred, not population-representative	Every performance claim must state the population it applies to.

12 · MONET concept scores — a sanity check that the data behaves



Mean MONET concept score per class, dermoscopic images. Colour = z-score across classes.

Read row by row: VASC scores highest on *vasculature/vessels*, INF on *erythema*, NV on *pigmented*. The concepts behave the way a dermatologist would expect, which is good evidence they carry real signal. Note also that *vasculature* averages 0.22 on dermoscopic images versus 0.02 on clinical ones — exactly what a dermatoscope is for.

Next session

Image representation and normalisation, then a first baseline — built on a grouped split, with malignant recall as the metric that counts.